A Deep Learning Approach to Stress Prediction Based on Sleep and Screen Time Patterns

Autor: Andrea Medina Rico **Student ID:** A01705541

Abstract. Here will be the abstract… soon :))

I. INTRODUCTION

Same as the abstract :))

II. DATA DESCRIPTION

This dataset is called *Sleep, Screen Time and Stress Analysis*, obtained from Kaggle. It contains 15, 000 synthetic observations that aim to represent individuals with different lifestyles and habits.

Categorical features:

- **user_id:** Unique identifier per observation.
- **gender:** Male, Female or Other.
- **occupation:** Professional background of the individual.

Numerical features:

- **age:** Age of the individuals.
- **daily_screen_time_hours**: Total of hours using digital devices.
- **phone_usage_before_sleep_minutes** Minutes using digital devices before going to sleep.
- **sleep_duration_hours**: Average hours of sleep per night.
- **sleep_quality_score**: Score from 1 to 10 ranking sleep quality.
- **stress_level**: Measured stress in scale from 1 to 10
- **caffeine_intake_cups**: Caffeinated drinks during the day.
- **physical_activity_minutes**: Minutes spent on physical activity.
- **notifications_received_per_day**: Notifications from smartphone per day.
- **mental_fatigue_score**: Mental fatigue in scale from1 to 10.

III. PREPROCESSING

a. Transformation

No null observations were identified, but a column does not help the purpose of the model, as it does not provide meaningful information about the individual's behavior: the user id. Therefore, the column was removed. Additionally, instances containing missing values were excluded, as their presence could bias the model and negatively affect predictive performance.

The other two categorical features were encoded in order to become an input to the model. The *gender* feature had three values, as shown previously. Due to the low representation of the *Other* category (4%) in *gender*, it was excluded to prevent the model from bias during training. This decision is based on data limitations and is not related to gender diversity considerations. The *gender* feature turned into a binary variable, having 0 for Male and 1 for Female.

Occupation had 8 different values: Designer, Teacher, Software Engineer, Manager, Student, Freelancer, Doctor and Researcher. To prevent the model from interpreting them as ordinal values, one hot encoding was applied. Using *k - 1* dummy variables, multicollinearity between them was avoided. The value *Designer* was not included, meaning when all the dummy variables equal 0, it corresponds to *Designer*.

Except for the scores and the stress level, the numerical features were obtained in two scales: hours and minutes. To ensure a consistent scale, all of them were converted to hours. Therefore, the column names were updated to reflect the correct units.

After the data split, all the features were standardized in order to improve training stability. It helps to give all the variables the same weight. This is done after the split to prevent data leakage from the test data.

b. Data correlation

The correlation matrix shows that most of the features have weak or almost null linear relationships between themselves and the target value. This indicates that stress level is influenced by complex relationships with the features rather than just linear. Therefore, a deep learning model could help capture this pattern.

There are only three features with a relatively high linear relationship: mental fatigue score, quality sleep score and daily screen time in hours. Considering the nature of the situation, it makes sense that these have a stronger relationship with stress.

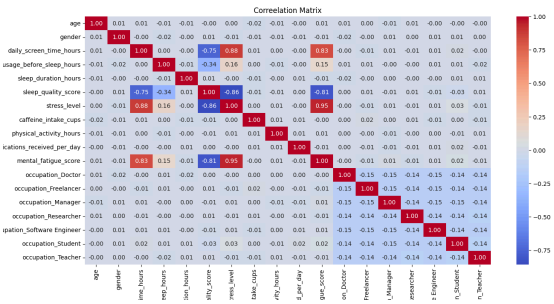

Figure 1. Correlation matrix

c. Data distribution

The target value stress level is skewed to the left with a higher concentration of observations near the maximum value. This also happens with the mental fatigue score.

The rest of features have either an almost perfect balance or a normal distribution.

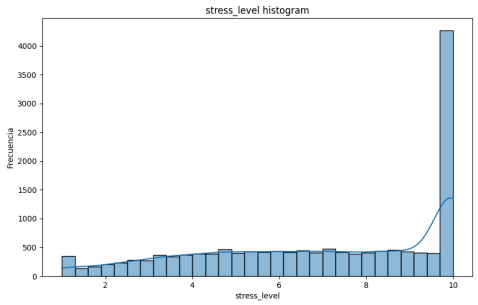

Figure 2. Stress level (target) histogram, skewed to the left.

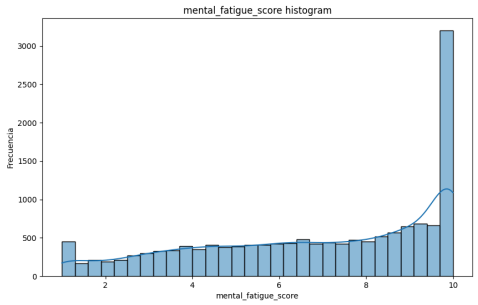

Figure 3. Mental fatigue score histogram, skewed to the left.

To improve training stability, all the features were standardized leaving them with mean equal 0 and standard deviation equal 1. This ensures that all variables contribute equally to the training. Features with a different distribution or scale don't influence more in the model. This process is done after the split, preventing data leakage from the testing instances.

d. Data split

The dataset does not have a previous separation for training and testing. Therefore, 70% of it will be dedicated to training, 15% to validation and 15% to test.

IV. MODEL

a. Justification

Prior work often formulates stress prediction as a classification problem.

However, in health-related contexts, treating the target as a continuous value through a regression approach allows for precise and informative predictions [1]. In this paper, several models were compared. It was proven that Multi-Layer Perceptron (MLP) can effectively identify complex dependencies in tabular health-related datasets, improving predictive performance compared to linear approaches

The loss function was Mean Squared Error (MSE), while the metric used for interpretation was Mean Absolute Error (MAE). These were chosen because MAE measures the average magnitude of prediction errors in the units of the target value, making it understandable for a human. MSE is used as a loss function as it penalizes larger deviations more strongly. Prior work in stress prediction problems has employed similar evaluation metrics, supporting their suitability for this task [2].

b. Model description

A MLP will be used with the following architecture:

- Dense layer 32 neurons
- Dense layer with 16 neurons
- Single-neuron output

The choice of a relatively simple architecture is intentional, as the dataset contains structured tabular data. This does not require very deep networks to be effectively modeled, reducing the risk of overfitting.

The first two layers use ReLU as their activation function to introduce non-linearity and capture more complex relations. The learning rate optimizer is Adam, which makes the learning rate adapt between epochs. It is suitable for problems with heterogeneous feature scales, common in behavioral datasets. The performance is measured in Mean Squared Error (MSE) and Mean Absolute Error (MAE).

c. Evaluation

As a diagnosis, the model is fit, meaning it doesn't present meaningful bias or difference between training and validation. The training and validation curves show a rapid decrease during the first epochs, indicating that the model quickly learns the patterns in the data. After approximately 5–10 epochs, both curves stabilize, suggesting convergence of the model. The final MAE suggests that the stress level predictions have an average error of 0.4748.

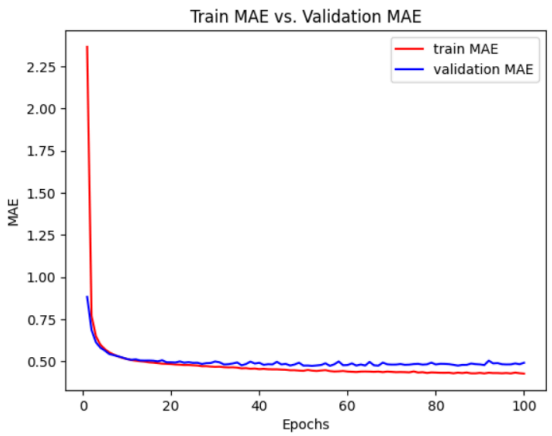

Figure 4. Train vs Validation MAE

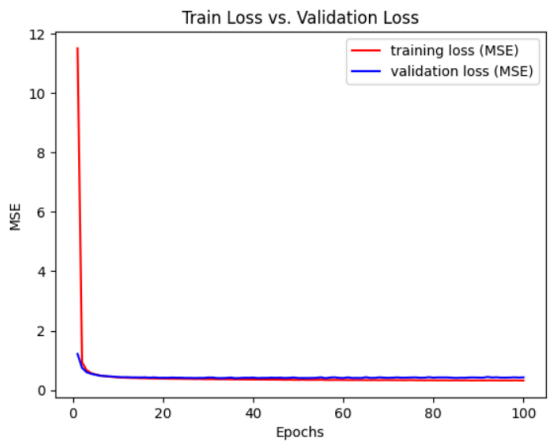

Figure 5. Train vs Validation Loss

The predicted versus actual values plot shows that most predictions are closely distributed around the ideal diagonal line

where $y = x$. However, there is a considerable amount of points deviating noticeably from the reference line, with the largest errors being approximately 2 to 3 units away from the true values.

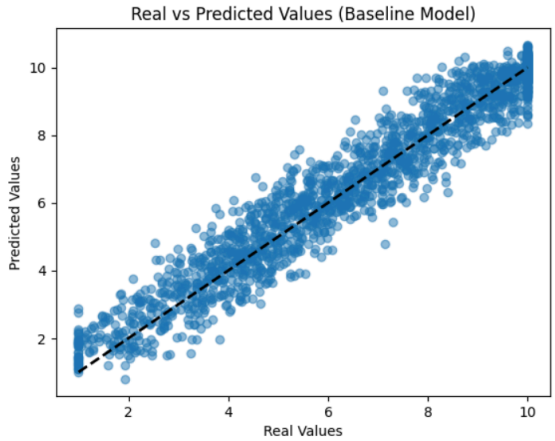

Figure 6. Real vs. Predicted Values

V. MODEL REFINEMENT

a. Model description

The refined architecture consists of:

- Dense with 128 neurons
- Dense with 64 neurons
- Dense with 32 neurons
- Single-neuron output

After the first two dense layers, Dropout was added. As the architecture became more complex and the dataset remains as structured tabular data, Dropout helps to prevent overfitting by randomly deactivating neurons during training. This encourages the model to learn more patterns and representations.

The activation function remains ReLU and the output layer remains linear to give the continuous value as a prediction.

The Adam optimizer is used with a reduced learning rate of 0.0005 to allow more stable parameter updates during training. This adjustment helps the model converge more smoothly and leads, as shown in the posterior graphs.

The model is trained for 150 epochs, increasing the number of iterations compared to the baseline to allow sufficient learning under the new architecture. Performance is evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE).

b. Evaluation

Even with the deeper architecture, the model does not present overfitting nor underfitting, keeping a similar performance in both train and validation. Adding more epochs helped the model to converge even more. The final MAE suggests that the stress level predictions have an average error of 0.4515.

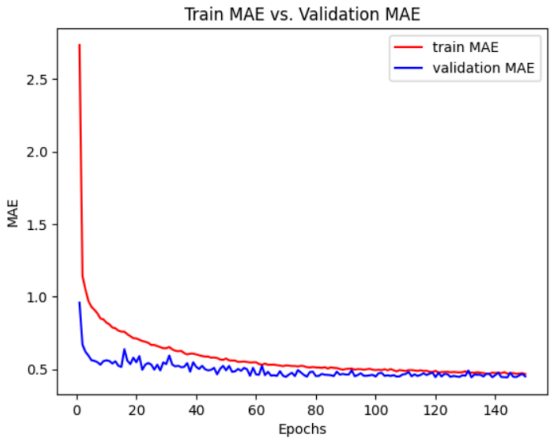

Figure 7. Train vs. Val MAE in refinement

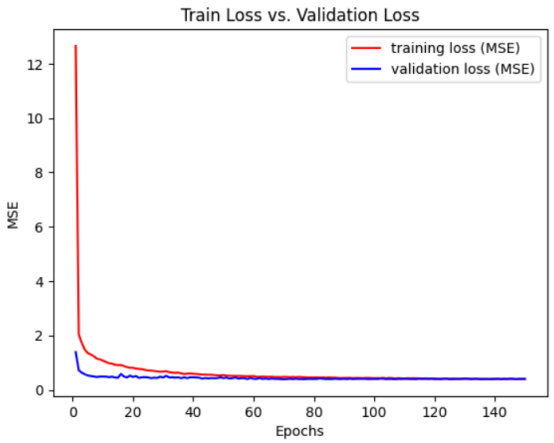

Figure 8. Train vs. Val Loss in refinement

The predicted values look closer to the line, as the color blue looks the darkest on the center. Even though some points are deviating, most of them are almost precise.

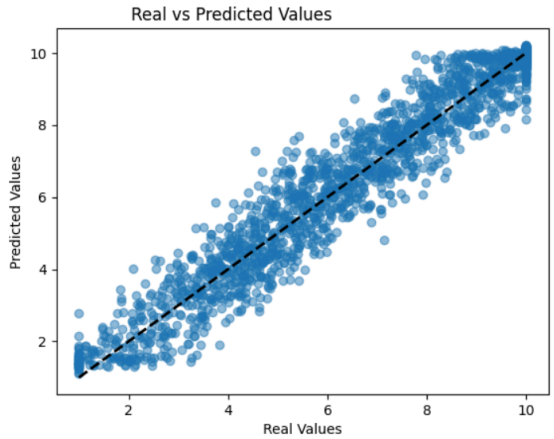

Figure 9. Real vs. Predicted values in refinement

VI. RESULTS COMPARISON

The refined architecture consists of three fully connected hidden layer

VII. CONCLUSION

The refined architecture consists of three fully connected hidden layer

A MLP will be used with the following architecture:

VIII. REFERENCES

[1] Alazaidah R, Samara G, Aljaidi M, Haj Qasem M, Alsarhan A, Alshammari M. Potential of Machine Learning for Predicting Sleep Disorders: A Comprehensive Analysis of Regression and Classification Models. Diagnostics (Basel). 2023 Dec 22;14(1):27. doi: 10.3390/diagnostics14010027. PMID: 38201336; PMCID: PMC10802836.

[2] Zhang Y, Qi E (2022) Happy work: Improving enterprise human resource management by predicting workers' stress using deep learning. PLOS ONE 17(4): e0266373. https://doi.org/10.1371/journal.pone.0266373